Harishwar Reddy K

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Homepage

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Google Scholar

Research Interests

Multi-modal data, Foundation Models, Self Supervised learning, Digital Pathology, Multi-Omics.

Education

University of Florida — Aug 2023 – May 2027(Anticipated)
Ph.D. in Electrical and Computer Engineering, CGPA: 4/4 — *Gainesville, Florida, USA*

Indian Institute of Science (IISc) — Oct 2020 – June 2022
M.Tech in Artificial Intelligence, CGPA: 8.2/10 — *Bangalore, Karnataka, India*

Selected Publications

- **Harishwar Reddy Kasireddy**, Patricio S. La Rosa, Akshita Gupta, Anindya S. Paul, Jamie L. Fermin, William L. Clapp, Meryl A. Waldman, Tarek M. El-Ashkar, Sanjay Jain, Luis Rodrigues, Kuang Yu Jen, Avi Z. Rosenberg, Michael T. Eadon, Jeffrey B. Hodgin, Pinaki Sarder, "A Comprehensive Benchmark of Histopathology Foundation Models for Kidney Digital Pathology Images", *Under Review in Nature Communications*

- Myles Joshua Toledo Tan, David A. Lichlyter, Michael M. Ebeid, Matthew W. Christopher, **Harishwar Reddy Kasireddy** *et al.*, "The Evolving Role of Machine Learning in Metabolomics for Precision Nutrition: A Systematic Review", *Under Review in IEEE JBHI*

- Nouar AlDahoul, Myles Joshua Tan, **Harishwar Reddy Kasireddy** Yasir Zaki, "Guardians of digital safety: benchmarking large language models in the fight against online toxicity ", In *Journal of Big Data*, 2026

- Nouar AlDahoul, Myles Joshua Toledo Tan,**Harishwar Reddy Kasireddy** Yasir Zaki "FaceScanPaliGemma multi-agent vision language models for facial attribute recognition ", *Nature Scientific Reports*, 2026

- **Harishwar Reddy Kasireddy**, Nicholas Lucarelli, Donghwan Yun, Kyung Chul Moon, Patricia S. La Rosa, *et al.*, "Explainable Feature Embeddings from Histopathology Foundation Models: A Case Study for End Stage Kidney Disease Risk Analysis in Diabetic Nephropathy Patients", In *Proceedings of Medical Imaging 2025: Digital and Computational Pathology*, 2025

- Myles Joshua Toledo Tan, **Harishwar Reddy Kasireddy**, Agus B. Satriya, H. Abdul Karim, Nouar AlDahoul, "Health Is Beyond Genetics: On the Integration of Lifestyle and Environment in Real-Time for Hyper-Personalized Medicine", *Frontiers in Public Health*, 12:1522673, 2025

- **Harishwar Reddy Kasireddy**, Udaykanth Reddy Kallam, Siddhartha M. S. K., Hemanth Kongara, Anshul Shivhare, Jayesh Saita, et al., "Deep-Learning-Based Visualization and Volumetric Analysis of Fluid Regions in Optical Coherence Tomography Scans", *Diagnostics*, 13(16):2659, 2023

- **Harishwar Reddy Kasireddy**, Anshul Shivhare, Hemanth Kongara, Jayesh Saita, Raghu Prasad, Chandra Sekhar Seelamantula, "Denoising Enhances Visualization of Optical Coherence Tomography Images", In *Proceedings of the Sixth NeurIPS Workshop on Medical Imaging (Med-NeurIPS)*, 2022

Research Experience

Graduate Research Assistant Aug 2023 – Present
University of Florida *Gainesville, FL, USA*

- Digital Pathology, Self Supervised learning (SSL), Spatial Transcriptomics, Segmentation, and Cell Detection.
- Trained the first multi-stain and multi-resolution histopathology foundation model for kidney whole slide images.
- Benchmarked histopathology foundation models for kidney digital pathology images.
- Trained a Multiple Instance Learning (MIL) model using the embeddings obtained from SSL model for drug treatment response prediction in Lupus Nephritis patients.
- Rigorous benchmarking of the SOTA histopathology foundation models for kidney specific histopathology tasks.
- Trained a SSL model for kidney specific histoapathology on 30,000 WSIs.
- Improving the explainability of histopathology foundation models, namely, UNI and Prov-Gigapath.
- Trained a glomeruli tuft segmentation model using only 400 glomeruli through Deeplabv3+ architecture
- Implemented Cell ViT for doing immune cell detection in PAS stained whole slide images (WSIs) for Lupus Nephritis patients.

Junior Research Fellow July 2022 – March 2023
Indian Institute of Science *Bangalore, India*

- Image Classification, Image Enhancement, and Visual Explanation of convolutional neural networks.
- Implemented Layer-CAM, Eigen-CAM, and HiResCAM on the Inception-ResNet-v2 architecture to classify abnormalities in Optical Coherence Tomography (OCT) data.
- Authored and submitted research papers to reputable conferences, including NeurIPS and ISBI.

Research Intern April 2023 – July 2023
Siemens *Bangalore, India*

- Explainable Active Learning (XAL), Deep Learning, MLflow, and Prefect.
- Fine-tuned models to classify Bags, Backpacks, and Suitcases in the Siemens logistics dataset.
- Developed an explainable active learning pipeline for the baggage dataset.

Skills

Technical Skills: Python, Pytorch, TensorFlow, OpenCV, NumPy, Pandas, Scikit-image, Scikit-learn, Matplotlib, 3D slicer, ImageJ, MATLAB, Qupath, ImageScope, MLflow, Prefect, Aperio, LaTeX.

References

- Dr. Pinaki Sarder, PhD
- Dr. Jeffrey B. Hodgin, MD, PhD
- Dr. Patricio S. La Rosa, PhD